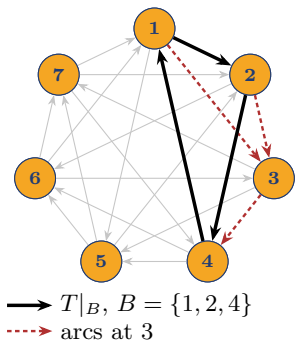


(a) the tournament P_7 

(b) the base state, forced

	0	1	2	3
π_1	1	2	4	
π_2	2	4	1	
π_3	4	1	2	

the only one

options, as slot triples:

 $D(3)$ 113 203 212 213 230 $|D(3)| = 5$ **fewest, so $v^* = 3$** $D(5)$ 032 122 131 132 302 $|D(5)| = 5$ $D(6)$ 023 221 311 320 321 $|D(6)| = 5$ $D(7)$ 000 001 002 003 010 011

012 020 030 100 101 110

111 120 200 201 300

 $|D(7)| = 17$

(d) an insertion of 3 that survives

put 3 at **230**:

π_1	1	2	3	4
π_2	2	4	1	3
π_3	3	4	1	2

π_1 still leads with (1, 2),
 and nothing has run
 out of room

what that leaves of (b)'s options:

 $D(5)$ 043 142 143 403 $|D(5)| = 4$ **was 5** $D(6)$ 024 222 420 $|D(6)| = 3$ **was 5** $D(7)$ 041 301 302 401 $|D(7)| = 4$ **was 17**(e) refining $D(6)$ past **230**:

<i>tuple</i>	<i>per voter</i>	<i>lift</i>	$c_6(3)$
023	=, =, +1	$\longrightarrow$ 024	2 ✓
221	split , =, +1	$\begin{matrix} \longrightarrow 222 \\ \searrow 322 \end{matrix}$	$\begin{matrix} 2 \text{ ✓} \\ 1 \text{ ✗} \end{matrix}$
311	+1, =, +1	$\longrightarrow$ 412	1 ✗
320	+1, =, split	$\begin{matrix} \longrightarrow 420 \\ \searrow 421 \end{matrix}$	$\begin{matrix} 2 \text{ ✓} \\ 1 \text{ ✗} \end{matrix}$
321	+1, =, +1	$\longrightarrow$ 422	1 ✗

(c) two rejected insertions of 3

put 3 at 113:

π_1	1	3	2	4
π_2	2	3	4	1
π_3	4	1	2	3

breaks Lemma 2.1: no
 voter can still lead with
 (1, 2) or (1, 4)

put 3 at 213:

π_1	1	2	3	4
π_2	2	3	4	1
π_3	4	1	2	3

$D(5 \mid B \cup \{3\}) = \emptyset$;
 and it makes $2 \rightarrow 3$
 unanimous

(f) the completed profile

π	1	2	3	4	5	6	7
$g\pi$	5	7	2	4	6	1	3
$g^2\pi$	6	3	7	4	1	5	2

$g = (156)(273)$, of order 3
 every arc at support 2